

Webnova 2026

webdevelopment hackethon

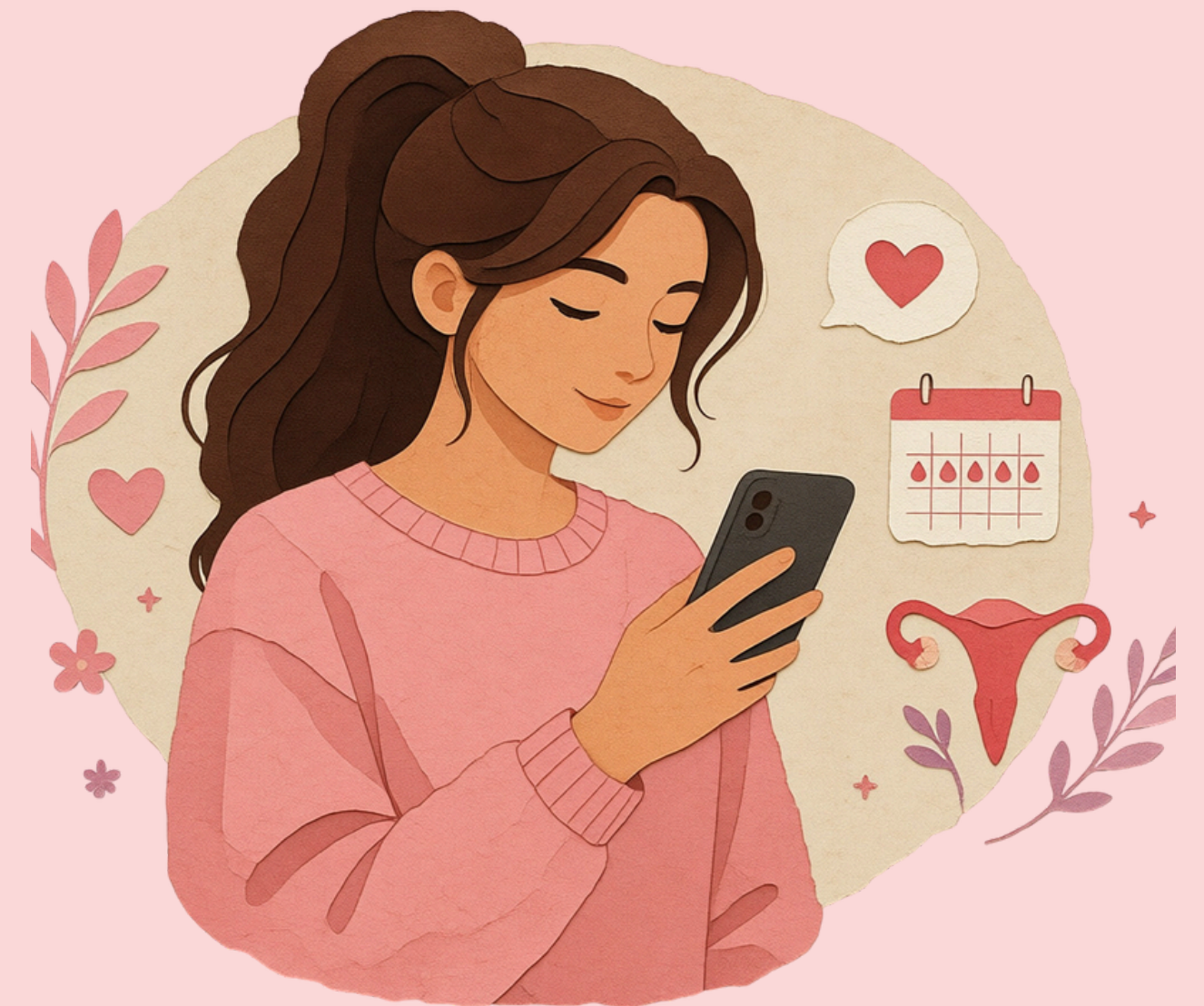
Team Name: Vibecoder

College Name: GL Bajaj, ABES

Engineering College

Team Members: Sneha, Chinky,vishal

Problem statement:healthcare



Problem Statement

What Real-World Problem Are We Solving?

Inaccurate Predictions – Existing apps struggle to track irregular menstrual cycles accurately.

Missed PCOD Signs – Early symptoms of PCOD are often ignored or undetected.

Lack of Monitoring – Women do not get continuous and personalized health tracking.

Delayed Consultation – Poor awareness and tracking can delay medical treatment and increase health risks.

who is affected & why it's matter

- **Teenagers with Early PCOD Symptoms** – Delayed detection can affect hormonal balance and future reproductive health.
- **Women with Irregular Menstrual Cycles** – Inaccurate cycle tracking creates difficulty in predicting periods and identifying health issues early.

Teenage girls
and women with
irregular cycles
Unpredictable patterns

Women at risk
of PCOD
Need early
detection tools

Working women
Limited time for
health monitoring

Doctors
Lack continuous
patient cycle data

1 in 5

Young women in India
affected by PCOS/PCOD

~70%

PCOS cases globally
go undiagnosed

Late

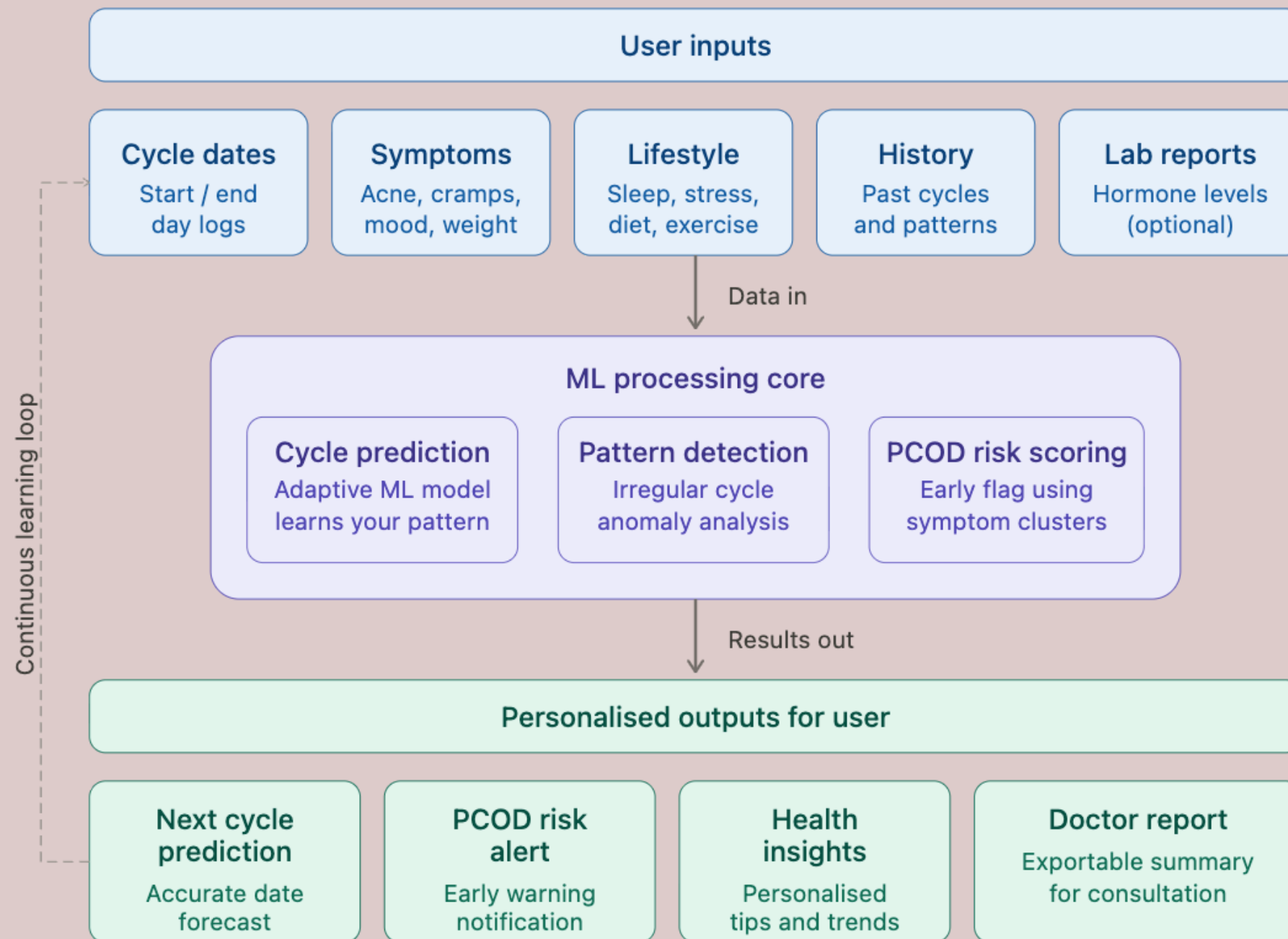
Typical diagnosis
often years after symptoms

who is affected

Women, especially teenagers and young adults with irregular menstrual cycles or PCOD/PCOS symptoms, are mainly affected due to lack of accurate health tracking and early detection

Proposed Solution

Proposed solution — AI-powered PCOD assistant



How our solution is different

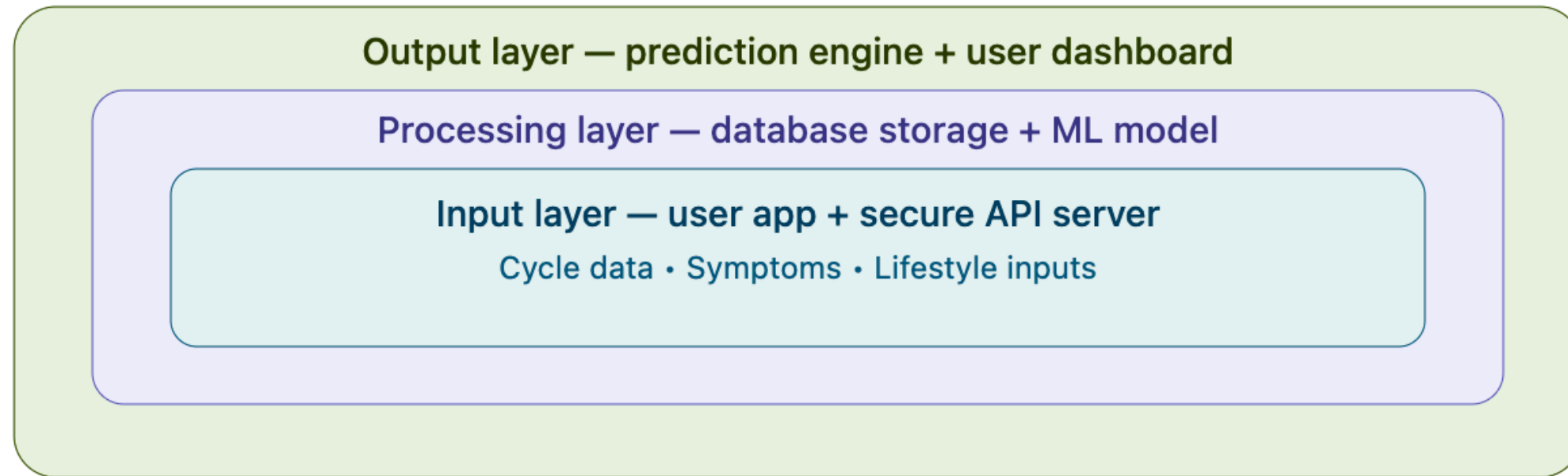


Irregular cycles, early PCOD risk, and doctor reports — all in one place.

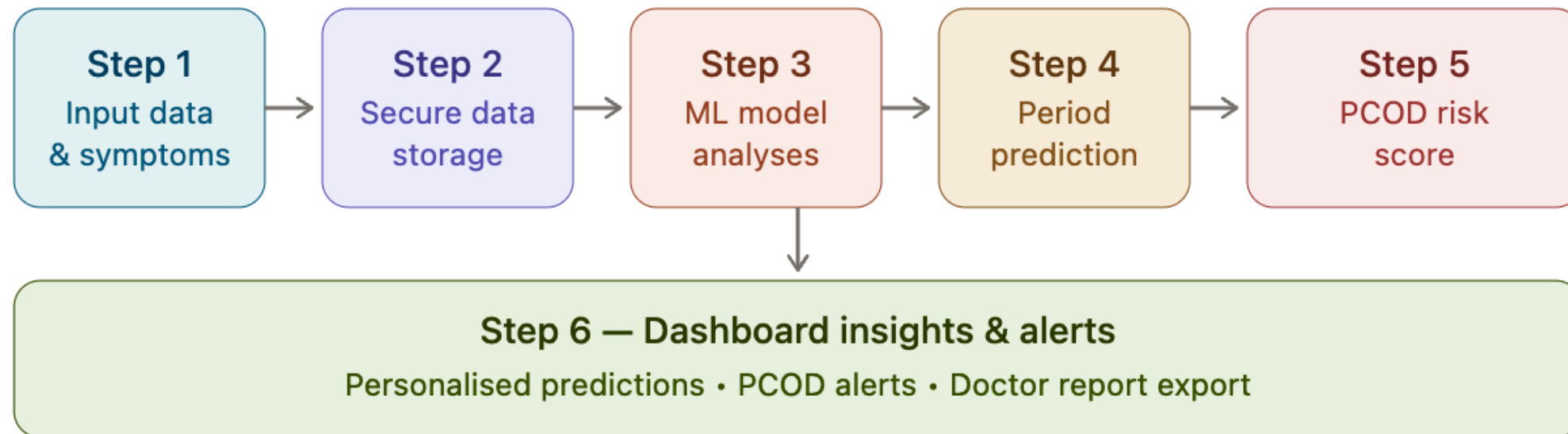
Technical Approach



System architecture



System working flow



key features

- **AI-Based Cycle Prediction** – Predicts menstrual cycles accurately even for irregular periods using smart analysis.
- **Early PCOD/PCOS Detection** – Identifies possible symptoms like delayed periods, acne, or hormonal imbalance at an early stage.
- **Personalized Health Tracking** – Tracks individual health patterns and provides customized insights and recommendations.
- **Period & Symptom Monitoring** – Monitors periods, ovulation, mood changes, pain, and other symptoms regularly.
- **Health Reminders & Alerts** – Sends reminders for periods, medication, water intake, and doctor consultations

IMPACT & BENEFITS

Built on evidence. Designed to change real health outcomes.



TARGET USERS

- Women with irregular cycles — 500M+ globally
- Undiagnosed PCOS patients — 70% of cases (WHO, 2023)
- Adolescents aged 13–18 — 1 in 5 young Indian women
- Gynaecologists needing structured patient cycle data



REAL-WORLD IMPACT

- Reactive treatment → Preventive cycle awareness
- Exportable reports cut the 3-doctor-visit burden
- Works on basic Android — no wearable or lab test needed
- Anonymized data can support national health policy
- Scalable to Tier 2/3 cities & underserved communities

70%

Undiagnosed

PCOS cases · WHO 2023

2+ Yrs

Avg Delay

Before diagnosis

85%

ML Accuracy

AUC · Pesapane et al. 2023



KEY BENEFITS

- ⚡ Cuts 2+ year PCOS diagnosis delay with early AI risk flagging
- 📄 Replaces 3 costly doctor visits with one shareable cycle report
- 🧠 Reduces anxiety from unpredictable, unexplained cycle patterns
- 🌐 Works on basic Android — no wearable or lab test needed
- 📈 Shifts women's health from reactive to preventive care at scale

"CycleAI shifts women's health from reactive treatment to preventive, personalized care."

RESEARCH & REFERENCES

Every claim in this presentation is backed by peer-reviewed research. We didn't guess — we built on evidence.



CLINICAL & EPIDEMIOLOGICAL RESEARCH

R1

Ganie et al. • JAMA Network Open • 2024

PCOS prevalence in India: 7.2%–19.6% across 9,824 women aged 18–40

R2

Dokras et al. • J. Clinical Endocrinology • 2017

33% of women wait 2+ years & see 3+ doctors before PCOS diagnosis

R3

WHO PCOS Fact Sheet • June 2023

8–13% of reproductive-age women have PCOS; up to 70% go undiagnosed

R4

Fernandez et al. • Human Reproduction • 2021 (PMC8289294)

Depression 50% higher in PCOS women; half of community cases undiagnosed



AI / ML RESEARCH + MARKET DATA

R5

Pesapane et al. • medRxiv • 2023

ML models (RF, SVM, GBT) predict PCOS before diagnosis — 80–85% AUC

R6

Zigarelli et al. • JMIR • 2022 (PMC8965679)

ML self-diagnostic tools for PCOS validated on real symptom + cycle data

R7

Umaa Mahesswari et al. • Heliyon • 2024 (PMC11530826)

Ensemble ML on cycle data achieves high-accuracy PCOS detection

R8

SNS Insider • Menstrual Health Apps Market • 2025

Market: \$1.72B (2024) → \$7.52B (2032) at 20.27% CAGR. Asia-Pacific fastest growing.

“Built on evidence. Designed for impact. Ready to scale.”